

H2BENCH · REQUIREMENTS

# H2Bench requirements

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DOCUMENT CONTROL

|             |                                      |         |                |
|-------------|--------------------------------------|---------|----------------|
| Document ID | HBN-REQ-001                          | Version | 0.2            |
| Status      | Draft                                | Date    | 2026-09-25     |
| Author      | Amish Chadha                         | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 2 Technology concept formulated  |         |                |
| Repository  | github.com/BoujeeEnjinia1701/h2bench |         |                |
| Source      | docs/03-requirements.md @ 5b7eff1    |         |                |

REVISION HISTORY

| Version | Date       | Author       | Change                                                           |
|---------|------------|--------------|------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with first-order status |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                 |

# H2Bench requirements

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These are first-pass requirements for the concept. Targets are proposals for review. The status column gives the first-order estimate from HBN-PRC-001; every "met" is an estimate to be checked by calculation at TRL 3 and by test later. Two requirements are **not met** (R9 and R14) and two are at risk (R8 and R10).

Table 1. Requirements.

| ID  | Requirement                                           | Target                                                                                                           | Status (estimate)                                                                                  | Verification (TRL 3 or later) |
|-----|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------|
| R1  | Measure the efficiency of each stage                  | Electrolyzer, storage, fuel cell and round trip, each to within $\pm 5$ % relative                               | Expected met: power to $\pm 1$ %, hydrogen quantity to about $\pm 2$ % by pressure and temperature | Uncertainty calculation (CAL) |
| R2  | Complete a full cycle in one lesson                   | Fill and discharge within 60 min of run time                                                                     | Met: about 18 min fill plus about 29 min discharge                                                 | Energy and flow calculation   |
| R3  | Electrolyzer power                                    | 100 W or less electrical input                                                                                   | Met: about 70 W                                                                                    | Stack datasheet               |
| R4  | Fuel cell power                                       | 50 W or less; 10 W or more net to run a visible load                                                             | Met: about 11 W net                                                                                | Stack datasheet               |
| R5  | Limit the hydrogen inventory                          | 10 L or less at 20 °C and 101.3 kPa                                                                              | Met: about 7.9 L (0.66 g) at the relief setting                                                    | Gas law calculation           |
| R6  | Limit storage pressure                                | Working pressure 300 kPa gauge or less; relief at 350 kPa gauge or less; vessel rated 1 MPa or more              | Met by design                                                                                      | Component ratings             |
| R7  | No flammable mixture in the room from a total release | Full inventory mixed into a 30 m <sup>3</sup> room below 25 % of the lower flammable limit                       | Met: about 0.03 % by volume, under 1 % of LFL                                                      | Calculation                   |
| R8  | Interlock shuts down on a leak                        | H2Guard alarm cuts electrolyzer power and closes the tank solenoid within 2 s; fan runs                          | At risk: depends on H2Guard, not yet specified                                                     | H2Guard design review         |
| R9  | Hydrogen quality at the fuel cell                     | Purity as the stack requires (the 12 W class specifies 99.995 % or better)                                       | <b>Not met on paper:</b> electrolyzer purity after the drier is not yet known                      | Supplier data, later gas test |
| R10 | Electrolyzer tolerates tank back-pressure             | H2 side rated to 300 kPa gauge or more over the O2 side                                                          | At risk: must be confirmed for the chosen stack                                                    | Supplier data                 |
| R11 | Water quality                                         | Conductivity 1 $\mu$ S/cm or less at the stack inlet, shown to students                                          | Met by design (mixed-bed resin and conductivity check)                                             | Datasheet                     |
| R12 | Log data students can analyze                         | Voltage, current, power, tank pressure and temperature at 1 Hz, saved as CSV                                     | Met by design                                                                                      | Firmware sketch review        |
| R13 | Fit an existing lab bench                             | Footprint 1,000 x 500 mm or less; height 800 mm or less; mass 25 kg or less                                      | Met: 900 x 450 mm, about 770 mm, about 18 kg (estimate)                                            | Massing model                 |
| R14 | Low cost and buildable                                | Parts cost \$450 or less, excluding H2Guard; hand tools only; no welding                                         | <b>Not met:</b> about \$845 (about \$780 without the bench supply)                                 | Priced BOM                    |
| R15 | Guard people from pressure parts                      | Tank inside a guard; relief and vent piped to the hood; no pressurized fitting that can be opened without a tool | Met by design                                                                                      | Design review                 |
| R16 |                                                       |                                                                                                                  | Expected met                                                                                       |                               |

| ID | Requirement                  | Target                                                            | Status (estimate) | Verification (TRL 3 or later) |
|----|------------------------------|-------------------------------------------------------------------|-------------------|-------------------------------|
|    | Set up and shut down quickly | Setup 15 min or less; tank vented and bench safe in 5 min or less |                   | Walk-through at TRL 3         |

## ■ Assumptions

- Electrolyzer: 4 cells at about 1.95 V and 9 A, Faraday efficiency 95 %; hydrogen higher heating value (HHV) 285.8 kJ/mol.
- Tank: 2.0 L internal, filled from 70 to 300 kPa gauge (171 to 401 kPa absolute) at 20 °C; below 70 kPa gauge the regulator cannot hold the fuel cell supply.
- Fuel cell: 13 cells at 0.60 V and 1.5 A, fuel utilization 95 % (the rest purged), about 0.9 W for fan and controls.
- Room: 30 m<sup>3</sup>, well mixed. The canopy hood exists because mixing near the leak is not instant; see HBN-PRC-001.
- Round-trip figures use the HHV basis. On a lower heating value (LHV) basis the hydrogen energy is about 15 % lower and the fuel cell efficiency correspondingly higher.